

03_HMI_DATA_LINEAGE

Document ID: DOC-02-ARCH-03 | Date: 2026-09-28 | Audited Standard: SIH Evaluator Evidence

EVIDENCE PROVENANCE: [CONTRACT VERIFICATION]

Forensic trace proving frontend dashboards are pure projection consumers with zero state fabrication (Rule 6).

1. Proof of Zero Client-Side Physics

In accordance with Rule 6, frontend applications (React 19 Control Room and In-Cab Driver Cockpits) are strictly decoupled from physics calculations. Every displayed value originates from the authoritative Digital Twin state store (twin_state_store.py) and is delivered over WebSocket.

2. Field-by-Field Data Lineage Matrix

HMI Element	Backend Data Source	Transport Channel	Provenance
Actual Speed (m/s)	twin_state_store.vehicles[id].speed_mps	WebSocket /ws/telemetry	PHYSICAL (DERIVED)
Wheel RPM	twin_state_store.vehicles[id].rpm	WebSocket /ws/telemetry	PHYSICAL (DERIVED)
Safe Speed Ceiling	twin_state_store.vehicles[id].safe_speed	WebSocket /ws/telemetry	PHYSICS SOLVER
Fog Visibility (m)	twin_state_store.environment.visibility_m	WebSocket /ws/telemetry	INJECTED WEATHER
Surface Friction	twin_state_store.roads[id].friction_mu	WebSocket /ws/telemetry	GEODATA PRIOR
Warning State	twin_state_store.vehicles[id].gov_state	WebSocket /ws/telemetry	SOFTWARE VERIFIED

3. In-Cab Warning Logic

- NORMAL (Green):** Actual speed <= safe speed. Cockpit renders green HUD indicator.
- CAUTION (Amber):** Actual speed approaching safe limit within 0.15 m/s.
- SLOW DOWN (Red):** Actual speed exceeds safe speed. Visual flash and audio chime instruct driver to retard.
- SEVERE STOP (Red):** Visibility < 30m. Driver instructed to hold at nearest designated passing bay.